

Nature's Recyclers: Rotting & Decomposition

A Detailed Lesson Plan for Grade 2

Subject: Environmental Science / General Science

Grade: Grade 2 (Ages 7–8)

Duration: 45–60 Minutes

Theme: Nature's Recyclers — Decomposition

Source: Planrs Academy

Lesson Overview

- Students will discover how living and once-living things break down naturally through the process of decomposition.
- They will meet the three main decomposers: bacteria, fungi, and earthworms.
- Students will sort materials into 'decomposes' and 'does not decompose' categories.
- Students will appreciate the ecological importance of decomposition and connect concepts to everyday Indian life.

Learning Objectives & Standards

1	Define decomposition in simple terms	Remember
2	Identify the three main types of decomposers (bacteria, fungi, earthworms)	Understand
3	Sort materials into biodegradable and non-biodegradable categories	Apply
4	Explain why decomposition is important for the environment	Analyse
5	Give local Indian examples of decomposition in everyday life	Apply
6	Distinguish myths from facts about decomposition and rotting	Evaluate

Key Vocabulary

Decomposition	The breaking down of dead matter into smaller parts by living organisms	
Decomposers	Living things that break down dead matter	Bacteria, fungi, earthworms
Bacteria	Microscopic single-celled organisms found everywhere	They make milk sour; breaks down food
Fungi	Organisms that absorb nutrients from dead matter	Mould on bread, mushrooms
Biodegradable	Materials that can be broken down naturally by decomposers	Decompose: paper, cotton, food
Non-biodegradable	Materials that cannot be broken down naturally	Plastic bags, glass, aluminium
Nutrients	Substances in soil that help plants grow	Nitrogen, phosphorus in soil

Materials & Preparation

Category	Item
Visual aids	Presentation slides / printed picture cards of decomposers
Sorting activity	Picture cards or real items: mango peel, banana peel, plastic bag,
Science journal	Blank paper or exercise book for each student
Classroom display	Chart paper and markers for vocabulary wall
Optional demo	A slice of bread left 2–3 days to show mould (in sealed bag)
Reference	Planrs Academy source slides

Lesson Structure at a Glance

1. Hook	Have You Seen This? — Discussion prompt	5 min
2. Explore	What is Decomposition? — Concept introduction	8 min
3. Meet	Decomposer characters: Bobby, Farida, Raja	10 min
4. Discuss	Why is decomposition important?	7 min
5. Sort	Sorting activity: biodegradable vs. non-biodegradable	8 min
6. Connect	Indian context — compost, banana leaves, dung	5 min
7. Correct	Bust the myths! Common misconceptions	5 min
8. Review	Vocabulary chant + key word recap	4 min
9. Close	Exit ticket / reflection	3 min
	Total	55 min

Detailed Lesson Phases

Phase 1: Hook — Have You Seen This? (5 min)

Opening Prompt:

Display a picture of a mouldy roti or overripe mango. Ask: '*Has anyone seen something like this at home? What do you think happened to it?*' Allow 2–3 students to share. Accept all answers warmly — there are no wrong observations!

Teacher Tips

- Use real or pictured local food examples for maximum connection.
- Validate student responses: 'Great observation!' builds confidence.
- Write the word DECOMPOSITION large on the board in green.

Phase 2: Explore — What is Decomposition? (8 min)

Core Explanation:

Introduce the definition: **Decomposition** is when something that was once alive breaks down slowly into tiny pieces and goes back into the soil. Emphasise that this is *Nature's Way of Recycling* — the Earth cleans itself!

Guide students to understand three key ideas:

- Dead things (leaves, food, wood) do not just disappear — they break down.
- Tiny living helpers do the work of breaking things apart.
- The broken-down matter returns to the soil as nutrients for new plants.

Phase 3: Meet the Decomposers (10 min)

Introduce the three characters one by one:

Bacteria Bobby — Tiny and Everywhere

- Too tiny to see without a microscope
- Found in soil, water, and air
- A million bacteria fit on one grain of rice
- Indian example: Bacteria turn milk sour (idli / curd fermentation!)

Fungi Farida — The Fuzzy Friend

- Grows as white, green, or black fuzzy patches
- Appears on bread, rotis, and fruit
- During monsoon, mould appears rapidly on stored food
- Mushrooms grow on rotting wood and cow dung

Worm Warrior — Kenchue Raja

- Lives in soil; eats dead leaves and makes soil rich
- After monsoon rains, earthworms come to the surface
- Termites also help break down old wood
- Earthworm castings are natural fertiliser

Phase 4: Why is Decomposition Important? (7 min)

Cleans the environment	Without decomposers, dead matter would pile up everywhere
Returns nutrients to soil	Plants need nutrients like nitrogen — decomposers release them
Helps plants grow	Rich soil from decomposition feeds crops and forests
Keeps ecosystems balanced	Every organism plays a role; decomposers close the nutrient cycle

Discussion question: 'What would happen if there were NO decomposers on Earth?' Allow a think-pair-share (30 seconds each). Expected answers: forests full of dead leaves, plants would run out of nutrients, soil would become poor.

Phase 5: Sorting Activity (8 min)

Give each pair of students a set of picture cards or real objects. Ask them to sort into two groups:

Decomposes (Biodegradable)

- Mango peel
- Banana peel
- Cooked food
- Dry leaves
- Marigold flowers
- Paper
- Cotton cloth

Does NOT Decompose (Non-biodegradable)

- Plastic bags
- Glass bottles
- Aluminium cans
- Synthetic fabrics
- Styrofoam cups
- Rubber tyres

Debrief Questions

- How did you decide which group each item belongs to?
- Why is plastic a problem for our environment?
- What can we do with biodegradable waste at home? (Compost!)

Phase 6: Indian Context — Nature's Helpers at Home (5 min)

Connect learning to local everyday experiences so students see science is *all around them*:

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Kitchen waste + garden leaves	Become rich compost for vegetable gardens
Banana leaves decompose quickly	Eco-friendly; used as plates in many Indian states
Cow dung (gobar)	Natural fertiliser; decomposes to enrich farm soil
Idli / curd fermentation	Bacteria doing beneficial work — same family as decomposers!
Earthworms after monsoon	Sign of healthy soil; nature's ploughs

Phase 7: Busting Myths! (5 min)

Myth	Fact
Rotting is bad and gross	Rotting = decomposition = nature's recycling — it's essential!
Only food rots	Leaves, wood, paper, cotton — all decompose too
All bacteria are harmful germs	Many bacteria are helpful; decomposer bacteria recycle nutrients
Decomposition happens instantly	It is a slow process — compost takes weeks to months

Phase 8: Vocabulary Review & Class Chant (4 min)

Decomposition	Breaking down of dead things
Bacteria	Tiny invisible helpers in the soil
Decomposers	Living things that break matter down
Fungi	Fuzzy decomposers — mould and mushrooms

Class Chant

Decomposers break it down, break it down, break it down!
Decomposers put it back in the ground!

Chant 3 times with clapping rhythm. Students can add actions: hands breaking apart (break it down), hands pressing to the ground (put it back).

Phase 9: Closure & Exit Ticket (3 min)

Exit Ticket Questions (choose one)

1. Name TWO decomposers and where you might find them.
2. Draw and label the decomposition cycle.
3. Write one sentence: 'Decomposition is important because...'
4. Give one Indian example of decomposition you have seen at home.

Assessment & Differentiation

Formative	Observe sorting activity; listen to think-pair-share	Checklist
Formative	Chant participation + verbal Q&A	Observation
Summative	Exit ticket written response	Written work
Summative	Draw & label the decomposition cycle	Drawing rubric

Learner Group

Support Strategy

Struggling learners	Provide sentence starters: 'Decomposition is...' Use picture-word cards for vocabulary
On-level learners	Standard activities as planned Peer discussion during sort
Advanced learners	Research a 4th decomposer (e.g., beetles, millipedes) Design a compost bin model
ELL / multilingual	Introduce terms in home language alongside English (e.g., Kenchue for earthworm)

Extension & Homework Activities

Compost Corner (Ongoing Project)

Set up a small classroom compost jar with kitchen scraps. Students observe and record changes over 2–3 weeks. Photograph each week to create a visual timeline.

Decomposer Hunt (Homework)

Students walk around their home or garden and find 3 examples of decomposition. Draw pictures and label what decomposer might be responsible.

Decomposer Diary (Creative Writing)

Students write a diary entry from the point of view of Bacteria Bobby or Fungi Farida describing their day 'at work' in the soil.

Sorting Game (Digital Extension)

Create a class chart: 'Things that decompose' vs 'Things that do not.' Students bring items from home to add to the chart each day for a week.

Guest Expert (Optional)

Invite a gardener or farmer (parent volunteer) to talk about composting or earthworms in real agricultural practice.

Teacher Background Notes

Decomposition is a biochemical process driven by microorganisms (primarily bacteria and fungi) and macro-organisms (earthworms, insects). These organisms secrete enzymes that break complex organic molecules (proteins, carbohydrates, lignin) into simpler inorganic compounds — a process called mineralisation. The released nutrients (especially nitrogen, phosphorus, and potassium) become available to plants through the soil solution. This cycle — often called the **nutrient cycle** or **decomposition loop** — is fundamental to all terrestrial ecosystems.

In Indian agricultural contexts, traditional composting (*compost khad*), vermicomposting (earthworm compost), and the use of cow dung (*gobar*) represent centuries of intuitive understanding of decomposition. Connecting classroom science to these cultural practices deepens both scientific and cultural appreciation.

Cross-Curricular Links

Language Arts	Vocabulary building; creative writing (Decomposer Diary)
Mathematics	Counting decomposers; measuring compost changes over time
Social Studies	Environmental responsibility; composting practices across India
Art	Illustrate the decomposition cycle; create decomposer character art
Health & Hygiene	Understanding helpful vs harmful bacteria; food safety

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